

Generation Interconnection System Impact Study Report

For Queue Project Q19-1214

Dooville 230 kV

June 27, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1214. The Interconnection Customer proposes a 319.2 MW solar photovoltaic facility interconnecting at the Dooville 230 kV substation in the Indiana area. The request was received on May 26, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 292 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,052,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Dooville / Thomas Hill Station 230.0 kV
Contingency: Dooville / Montpelier Electric Generating Sta 230.0 kV line
Loading: 121.1% (Rating: 925 MVA)
DFAX: 100.0%
Pre-project loading: 86.6%
2. Facility: Murdock / Sabanera 230.0 kV
Contingency: E Methuen / Henry County 500.0 kV
Loading: 100.2% (Rating: 731 MVA)
DFAX: 7.8%
Pre-project loading: 96.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Dooville / Thomas Hill Station 230.0 kV	Rebuild line with higher-capacity conductor	\$34,892,000	28% (\$9,706,000)
Murdock / Sabanera 230.0 kV	Reconductor line	\$10,514,000	3% (\$346,000)
	Total allocated to project		\$10,052,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.